UNIT - 1

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Concept of Watershed | Introduction to Watershed Management | Stakeholders and relative importance | Policies and decision making | Practices in arid & semi-arid regions | Short term & long term strategic planning | Types & sources of pollution | Environmental guidelines for water quality | Perspective on recycle and reuse

1. Concept of Watershed

Watershed: the area of land from which all rain water (surface runoff) drains to a common outlet such as a stream, river, lake, reservoir or the sea. It is bounded by a high ridge line called the water divide.

Other names: catchment, drainage basin, river basin (in the USA: watershed; in the UK: catchment).

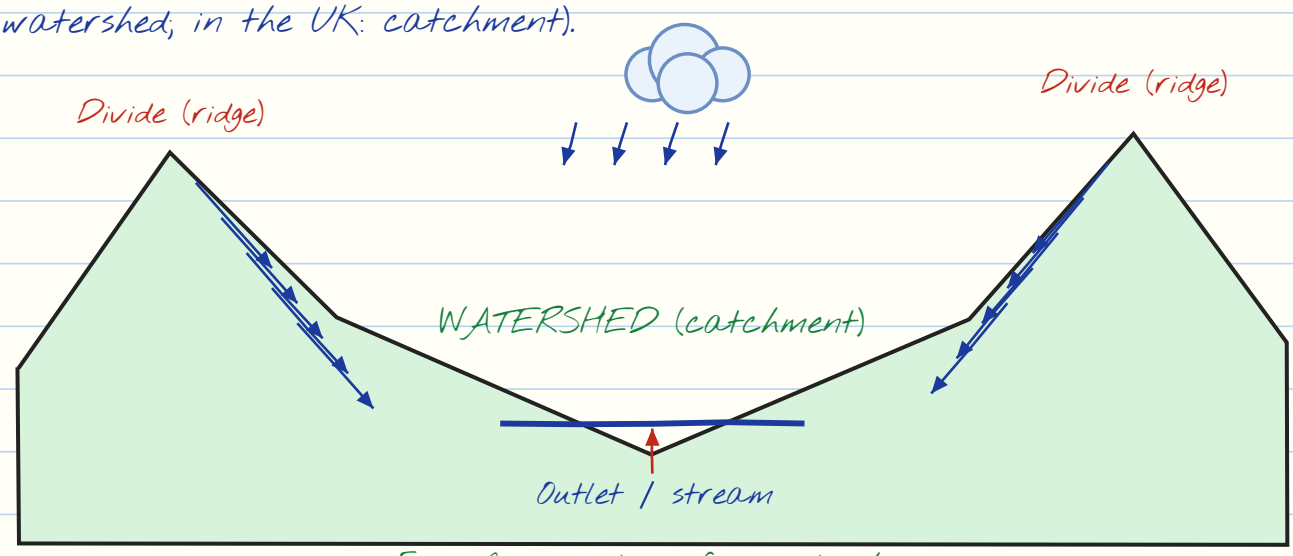


Fig 1: Cross-section of a watershed

Components / Characteristics

* **Divide:** the ridge line that separates two watersheds. Rain falling on either side flows to a different outlet.

* Outlet (pour point): the lowest point where all runoff leaves the watershed.
* Drainage network: main stream + tributaries (streams are ordered 1st, 2nd, 3rd ... by Strahler method).
* Land use: forest, agriculture, grazing land, wasteland, settlements. Land use controls runoff and erosion.
* Physical features: area, shape, slope, relief, soil type, geology, vegetation cover.

Classification by size (AIS and LUS, India)



Fig 2: Hierarchy of watersheds (by area)

* Hydrologically each watershed behaves as an open system: Input = rainfall; Output = runoff, ET, percolation, storage change.

$$P = Q + ET + dS$$ (water balance)

where P = precipitation, Q = runoff, ET = evapotranspiration, dS = change in storage (soil moisture + groundwater).

Why the watershed is the best planning unit?

1. Natural hydrological unit: effect of any action upstream (land use, dam, pollution) is felt downstream at the outlet.
2. Links land, water, vegetation and people in one system - an integrated approach is possible.
3. Upstream-downstream linkages can be studied: erosion upstream means siltation downstream.

4. Easy to monitor: measuring flow and sediment at one outlet gives the effect of all treatments.
5. Suitable for community participation because a micro-watershed generally covers 1 to 3 villages.

2. Introduction to Watershed Management

Watershed Management: the process of formulating and carrying out a course of action involving manipulation of the natural system of a watershed to achieve specific objectives - while protecting soil, water and vegetation and improving the livelihood of people living in it.

> Definition in short:
> Watershed management is NOT just soil and water conservation. It is the integrated use of land, water and vegetation (and also human and livestock resources) on a drainage-basin basis for optimum and sustained production with minimum hazard to natural resources.

Objectives

* Control runoff and soil erosion; conserve rain water in situ (where it falls).
* Increase infiltration, recharge groundwater, raise water table in wells.
* Improve soil fertility and increase crop, fodder, fuel and timber production.
* Reduce siltation of tanks, reservoirs and flood peaks; protect downstream areas.
* Ensure sustainable livelihood, reduce poverty, create employment in rural areas.
* Maintain ecological balance - biodiversity, water quality, climate resilience.

Principles of watershed management

* Utilise land according to its capability (land capability classes I to VIII).
* Protect fertile land from degradation and put every field under vegetative cover.
* Conserve as much rain water as possible where it falls (in situ).
* Drain out excess water safely at non-erosive velocity.
* Store excess water in ponds/check dams for supplemental irrigation.
* Treat the watershed ridge-to-valley: upper reach -> middle reach -> lower reach.
* Build capacity of people, make users the owners and managers of the work.

Scope / Components of the programme

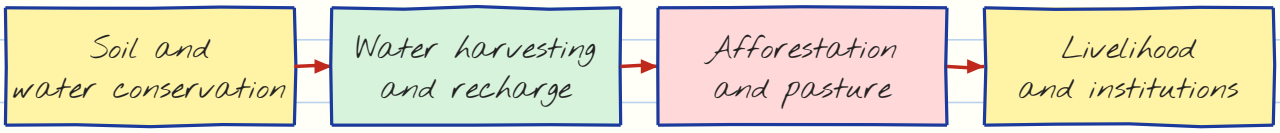


Fig 3: Main components

Component	Typical measures
Land treatment	contour bunds, terraces, trenches, gully plugs, vegetative barriers
Drainage line	check dams, loose boulder structures, gabions, farm ponds
Biomass	agro-forestry, silvi-pasture, horticulture, fodder
Livelihood	SHGs, dairy, micro-enterprise, training

3. Different Stakeholders and Their Relative Importance

Stakeholder: any person, group or institution that affects, or is affected by, the watershed and its management.

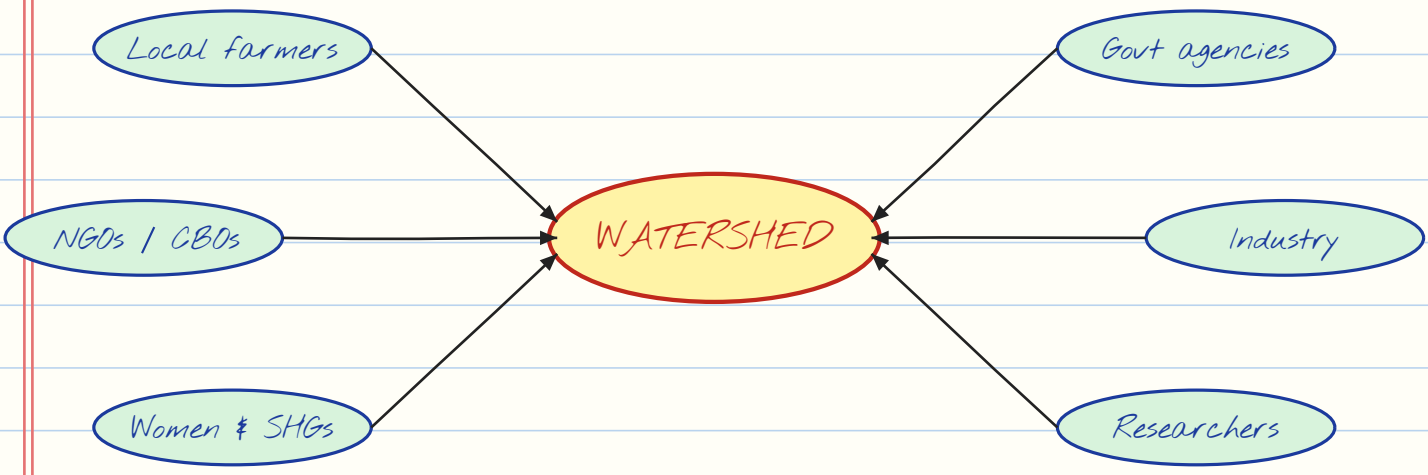


Fig 4: Stakeholders around a watershed

Main groups

* Primary stakeholders: farmers, landless labour, women, livestock keepers, tribal and forest dwellers - directly depend on the land and water; they bear the cost and get the benefits.
* Secondary stakeholders: Gram Panchayat, line departments (agriculture, forest, irrigation, rural development), banks, NGOs, research institutes, extension agencies.
* Tertiary / external: downstream users, urban water supply, industries, tourists, policy makers, funding agencies.

Relative importance (influence - interest)

Group	Interest	Influence	Role / need
Small farmers, women	Very high	Low (needs support)	Primary users, to be empowered and given a voice
Large landholders	High	High	Can block or help, benefits must be shared fairly
Government depts	Medium	Very high	Funds, technology, rules, monitoring
NGOs / CBOs	High	Medium	Facilitate, mobilise, train, bridge gap
Industry / urban users	Medium	High	Water demand, pollution, can fund CSR
Researchers	Medium	Low-Med	Data, models, impact assessment

Remember:

Equity rule: priority should go to the high-interest-low-influence group (poor, women, landless). If they are ignored, the project fails after the funding ends. Stakeholder analysis = identify -> map -> engage -> resolve conflicts.

Conflicts arise: upstream vs downstream, farmers vs graziers, landed vs landless. Solved through Watershed Committee, User Groups, transparent social audit and benefit sharing.

4. Watershed Management Policies and Decision Making

Policy framework in India

* National Water Policy (1987, 2002, 2012): water as a national resource, river basin as unit, priority to drinking water, then irrigation, hydropower, industry.
* Integrated Watershed Management Programme (IWMP, 2009-10): merged DPAP, DDP, IWDP, later in PMKSY - Watershed Development component (2015).

* Common Guidelines for Watershed Development Projects 2008 (revised 2011): cluster approach, 4-7 year project, 5-8 per cent for capacity building, 1000 ha per project minimum.
* Other: MGNREGA (works on water conservation), National Mission for Sustainable Agriculture, Jal Shakti Abhiyan, Atal Bhujal Yojana, Mission Amrit Sarovar.

Decision-making in watershed management

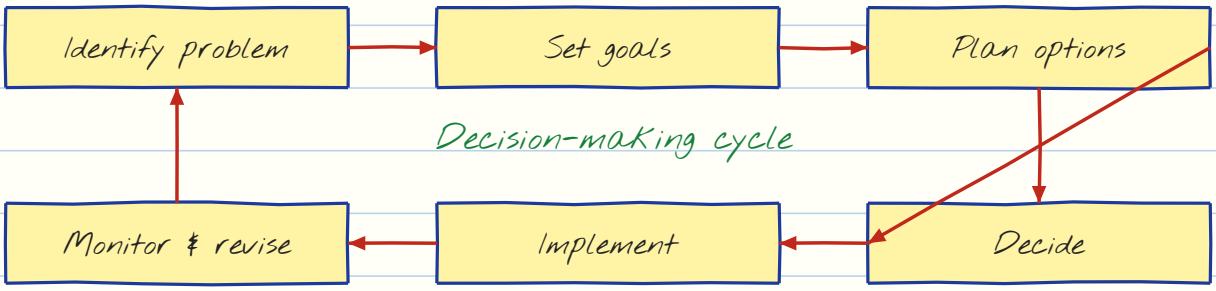


Fig 5: Participatory decision cycle

* Bottom-up: PRA (Participatory Rural Appraisal) with villagers decides problems and priorities.
* Tools: GIS and remote sensing, hydrological models, multi-criteria analysis, cost-benefit analysis, stakeholder consultation.
* Criteria for priority: runoff potential, erosion class, slope, land use, poverty, drinking-water scarcity, groundwater status.
* Adaptive management: monitor, learn and change decision with new information.

> Principle:
> Good decisions are: participatory, transparent, science-based, equitable, economically viable and environmentally sustainable.

5. Practices in Arid and Semi-arid Regions

Arid: rainfall below 250-400 mm/yr, very high PET; Semi-arid: rainfall 400-800 mm with erratic distribution, long dry spells and frequent droughts. About 2/3 of India's cultivated area is rainfed.

Problems

* Low and highly variable rainfall; high intensity storms cause erosion.
* Shallow, coarse, low-organic-matter soils; crusting; wind erosion in deserts.
* Low groundwater recharge, salinity, over-extraction; drought-prone livelihoods.

Management practices

* In situ moisture conservation: contour bunding, graded bunds, compartment bunds, contour trenching, broad-bed and furrow, tied ridges, mulching, conservation tillage.
* Water harvesting: farm ponds, nala bunds, check dams, percolation tanks, Khadin (Rajasthan), tankas, johads, ahar-pyne, tanka.
* Wind erosion control: shelter belts, wind breaks, sand-dune stabilisation using grasses (Cenchrus, Lasiurus) and trees (Prosopis, Acacia).
* Drought-tolerant crops: bajra, jowar, pulses, castor; contingency crop plans; short-duration varieties.
* Silvi-pasture and agro-forestry: Hardwickia, Prosopis cineraria (Khejri), Ziziphus (ber), Azadirachta (neem).
* Efficient irrigation: drip, sprinkler, protective irrigation, supplemental irrigation at critical stages.
* Contingency: fodder banks, crop insurance, grain banks, community seed banks.

> Slogan:
> Rule for dry regions: 'Catch the rain where it falls, store it, use it efficiently.'

6. Short Term and Long Term Strategic Planning

Aspect	Short term (1-3 yrs)	Long term (5-25 yrs)
Goal	quick visible benefits, build trust	sustainable productivity, ecology
Actions	gully plugs, field bunds, farm ponds, drinking water, nursery	afforestation, terracing, groundwater recharge, institutions, markets
Finance	project grants, wage labour	watershed development fund, user charges, bank credit
Output	reduced runoff, more soil moisture	raised water table, stable yields, biodiversity

* Strategic planning steps: (1) vision and objectives, (2) baseline survey and resource inventory, (3) problem analysis, (4) alternatives and priorities, (5) action plan, budget, schedule, (6) implementation, (7) monitoring and evaluation (M and E), (8) exit and sustainability.
* Phases in watershed projects: preparatory (3 yrs) -> works phase (3-4 yrs) -> consolidation / withdrawal phase.
* Short-term wins keep people interested; long-term plans safeguard the resource base. Both are linked through a Detailed Project Report (DPR).

7. Types and Sources of Pollution

Water pollution: addition of unwanted substances (physical, chemical, biological) which make water unfit for its intended use.

Point sources (pipes) vs Non-point sources (runoff)

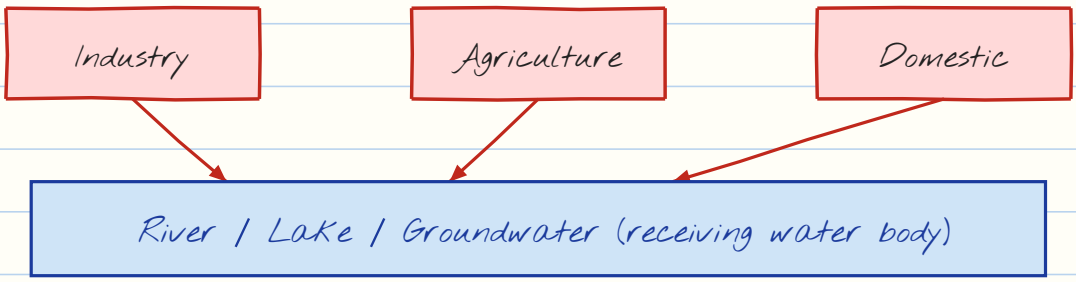


Fig 6: Pollution sources

Sources

* Point source: from a single identifiable location - industrial effluent pipe, sewage outfall, mine drainage. Easy to monitor, can be treated.
* Non-point (diffuse) source: no single point - agricultural runoff (fertiliser, pesticides), urban storm water, soil erosion. Hard to control.

Types of pollutants

Type	Examples	Effect
Organic	sewage, food waste, paper mills	high BOD, low DO, fish kill
Nutrients	nitrogen and phosphorus from fertiliser	eutrophication, algal bloom
Pathogens	bacteria, viruses, protozoa	cholera, typhoid, hepatitis
Sediment	eroded soil	turbidity, reservoir siltation
Toxic / heavy metals	Pb, Hg, Cd, As, pesticides	bio-accumulation, cancer
Thermal	power plant cooling water	reduced DO, ecosystem change
Oil / plastics / emerging	oil spills, microplastics, pharmaceuticals	toxic to aquatic life

* Watershed link: erosion + fertiliser + livestock waste + poor sanitation in the catchment decide the quality of water at the outlet.

8. Environmental Guidelines for Water Quality

Standards give the permissible limits of each parameter for a given use.

Common bodies: BIS (IS 10500), CPCB, WHO, US-EPA.

Parameter	Drinking (IS 10500)	Remarks
pH	6.5 – 8.5	acid/alkaline limit
TDS	500 mg/L (2000 if no alternative)	taste, salinity
Turbidity	1 NTU (5 permissible)	pathogens hide in particles
Hardness	200 mg/L (600)	as CaCO3
Nitrate	45 mg/L	blue baby disease
Fluoride	1.0 mg/L (1.5)	fluorosis
Arsenic	0.01 mg/L	toxic, cancer
Coliform	absent in 100 mL	faecal contamination

* CPCB classes of surface water: A (drinking after disinfection, DO > 6, BOD < 2), B (outdoor bathing, BOD < 3), C (drinking after conventional treatment, BOD < 3), D (fish and wildlife, DO > 4), E (irrigation, industrial cooling).
* Irrigation water: EC < 2 dS/m, SAR (sodium adsorption ratio) < 10 for safe use; salinity and sodium hazard classes of US Salinity Lab.
* Environmental flow: minimum flow to keep aquatic ecosystem healthy must be left in streams.
* Effluent discharge standards: BOD <= 30 mg/L for inland surface waters (CPCB general standard), COD <= 250 mg/L, TSS <= 100 mg/L.

BOD = (DO initial − DO after 5 days at 20 C) × dilution factor

9. Perspective on Recycle and Reuse

Scarcity and rising demand make water a resource that must be used again and again – 'reduce, reuse, recycle, recharge'.

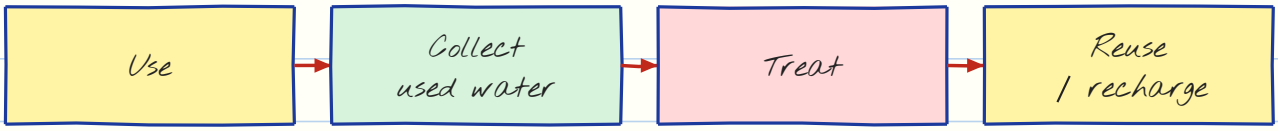


Fig 7: Reuse loop

* Reuse: using wastewater (with or without treatment) again for another purpose, e.g. treated sewage for gardens, toilets, construction, cooling.
* Recycle: treated water used again for the same purpose in a closed loop, e.g. industrial process water.
* Greywater (bath, laundry, kitchen) -> toilet flushing, gardening. Blackwater needs full treatment.
* Treated municipal wastewater (STP, constructed wetlands) -> irrigation of non-food/fodder/treecrops, industrial cooling, groundwater recharge.
* Farm level: tail-water recovery, return flow, use of drainage water in conjunctive way with fresh water.
* Rain water harvesting is also 'reuse' of natural precipitation.

Benefits and concerns

* Benefits: reduces fresh water demand and pollution load, cheap supply, nutrients (N, P) for crops, drought-proof source.
* Concerns: pathogens, salts, heavy metals, public acceptance, needs treatment standards, monitoring and health safeguards.

> Take-away:
> Wastewater is a resource, not a waste. Safe reuse = right treatment + right crop/use + right application method + monitoring (WHO multi-barrier guidelines).

Examples: Singapore NEWater, Israel (reuses > 85% of its sewage for agriculture), Nagpur STP water for power plant, Surat treated water for industries.